Task

The Argumentation Mining task (AM) aims to identify argumentative structures in texts. Argumentation Mining mainly includes 4 subtasks:

Argumentation Component Identification: Separate argumentation components from unstructured text.

Argumentation Component Type Classification: classify their categories, such as premise, claim, etc.

Argumentation Relation Identification: identify whether there is an argumentative relationship between the argumentation components.

Argumentation Relation Type Classification: Classify the relations between argumentation components, i.e. support, attack, etc.

We focus the last 3 subtasks.

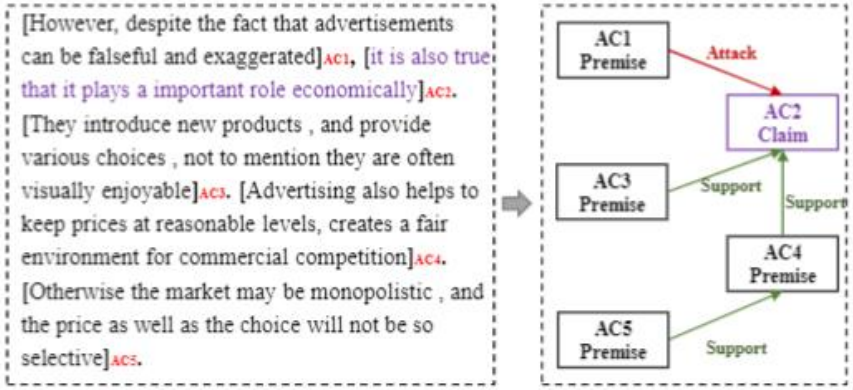


Fig. 1. An example of argumentation copus form the PE dataset, where *Premise* and *Claim* denote the type of ACs. Solid directed lines denote ARs. *Attack* and *Support* refer to the types of ARs.

Motivation

Challenges that are not addressed well in the literature:

(i) previous methods tend to apply mean pooling or weighted sum operation to learn the representations of ACs, i.e., coarse-grained AC representations (Morio et al.,2020; Bao et al., 2021).

(ii) previous works cannot solve these three subtasks in a unified framework on both the tree (Stab and Gurevych, 2014) and non-tree (Park and Cardie, 2018) structured argumentation. Although there are several studies (Kuribayashi et al., 2019; Morio,Ozaki, Morishita, Koreeda and Yanai, 2020) attemptingto solve ARI, ACTC and ARTC jointly, they only focuson either tree or non-tree structure (but not on both).

Contributions:

(i) Tackle Argumentation Mining (AM) task from various perspectives by constructing a graph that represents relations among the argument components (ACs).

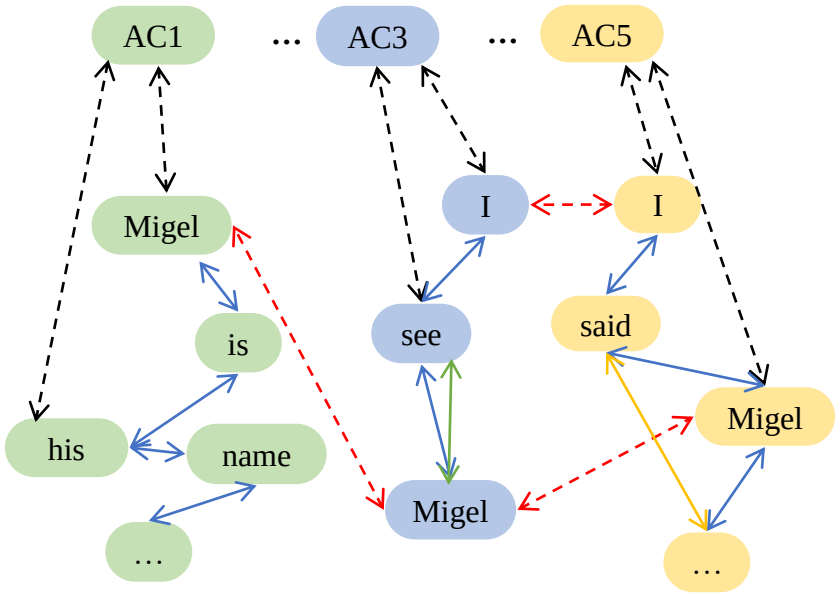
(ii) explicitly parses the significant relations among ACs, which fully exploits the rich and complementary information between the subtasks for modeling the argumentation structures.

(iii) Resolving the three interacted subtasks of AM (i.e., ACTC, ARI and ARCT) in a unified multi-task learning framework on both tree and non-tree structured argumentation.

Methods

Intra- and Inter- Relation Graph & EEGCN

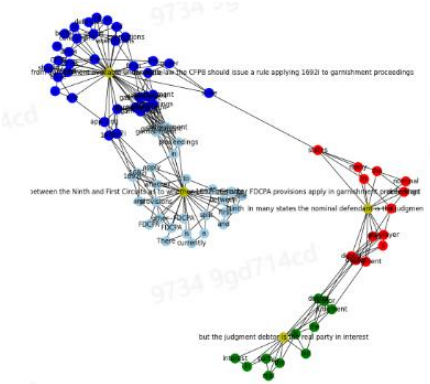
Argumentative text can form a graph that consists of both intra and inter relations.



Relations:

- Parent_to_Child
- Adjacency (intra)
- Dependency (intra)
- Lexical Consistency (inter)
- Coreference (inter)

Explicitly mine the relationship between words in same AC (intra) and the relationship between words in different ACs (inter), and build a relationship graph. Using two types of relationships can better capture the information of AC, so as to complete the modeling.

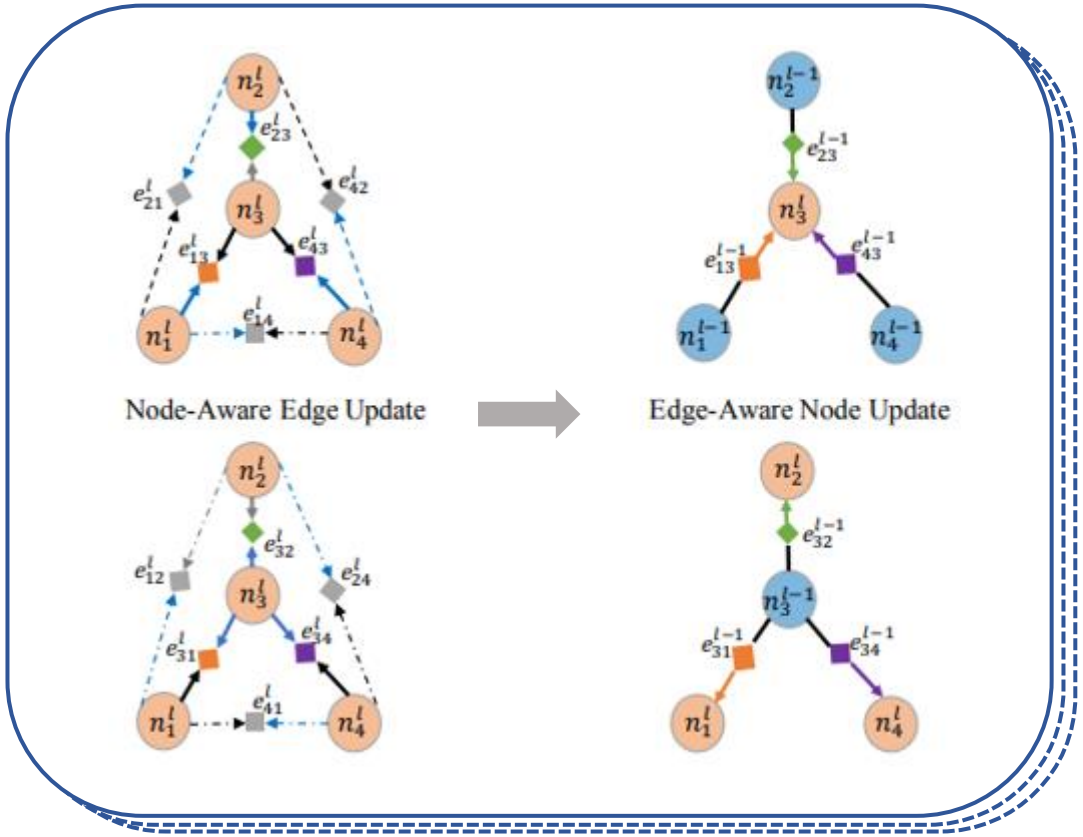


real graph constructed from a sample

Methods

Intra- and Inter- Relation Graph & EEGCN

use Eadg-Enhanced GCN to encoding the graph.



L-layers

Based on GCN:

1. Node-Aware Edge Update

$$\mathbf{E}^l_{i,j,:} = \mathrm{NAEU}(\mathbf{E}^{l-1}_{i,j,:}, \mathbf{h}^l_i, \mathbf{h}^l_j)$$
$$= \mathbf{W}_u[\mathbf{E}^{l-1}_{i,j,:} \oplus \mathbf{h}^l_i \oplus \mathbf{h}^l_j], i, j \in [1, n],$$

2. Edge-Aware Node Update

$$\mathbf{H}^l = \mathrm{EANU}(\mathbf{E}^{l-1}, \mathbf{H}^{l-1})$$
$$= \sigma(\mathrm{Pool}(\mathbf{H}^l_1, \mathbf{H}^l_2, ..., \mathbf{H}^l_p)). \quad \mathbf{H}^l_i = \mathbf{E}^{l-1}_{:,:,i}\mathbf{H}^{l-1}\mathbf{W},$$

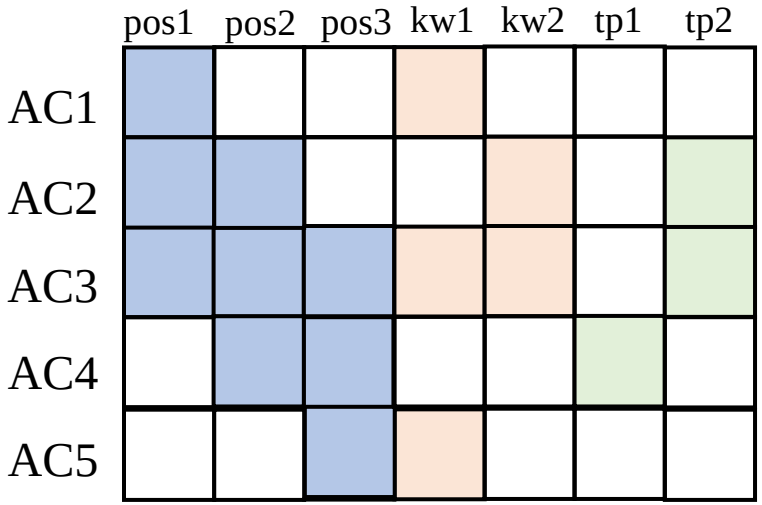
The performance is under SOTA:

1. Intra & inter graph didn't contain logic reasoning knowledge
2. Relations(that we want to investigate) between ACs is indirect, must cross each other's children. EEGCN is simple and not good at it.
3. EEGCN cannot handle the situation that 2 or more edges between 2 nodes.

Methods

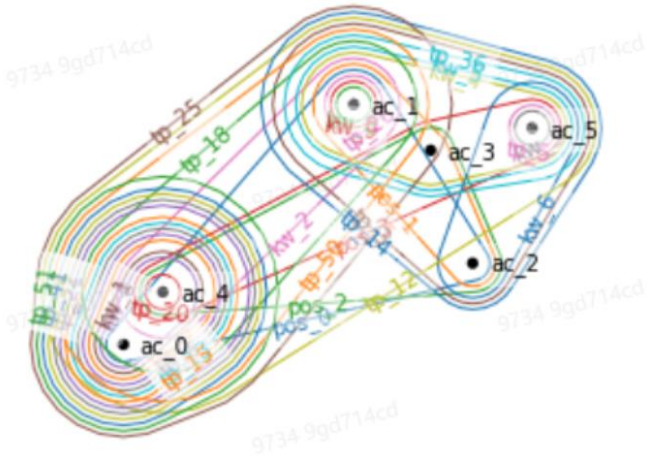
HyperGraph & HGTrans

Argumentative text can also form a hypergraph.



Relations:

- Position (local)
- Keyword (local)
- Topic (global)

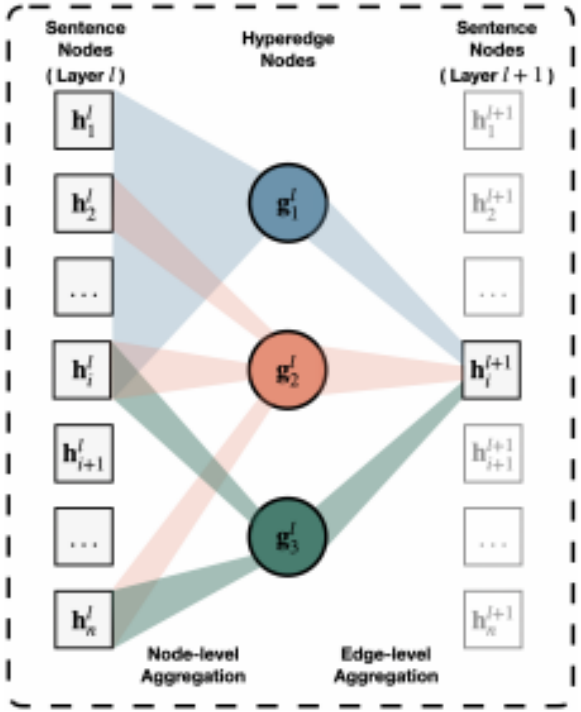


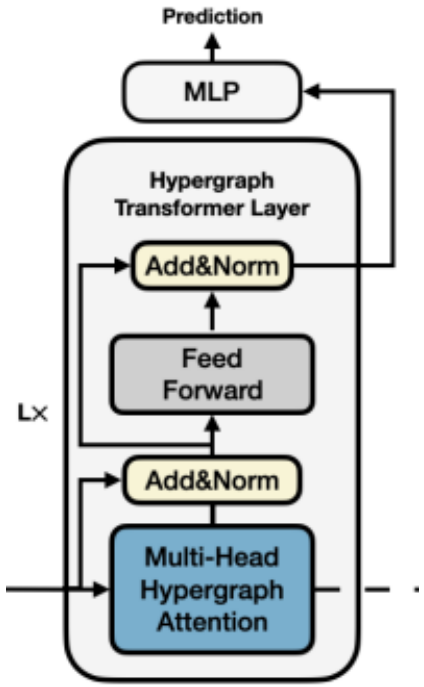
real graph constructed from a sample

A hyperedge can connect more than 2 nodes in hypergraph.
We investigate both local and global relations among ACs.

Methods

Inspired by Attention Scheme and Tranformer Architecture, using HyperGraphTransformer.





Based on Attention:

1. Node-Aware Edge Update

$$\mathbf{g}_j^l = \mathrm{LeakyReLU}\left(\sum_{v_k \in e_j} \alpha_{jk} \mathbf{W}_h \mathbf{h}_k^{l-1}\right),$$

$$\alpha_{jk} = \frac{\exp\left(\mathbf{w}_{ah}^{\mathrm{T}} \mathbf{u}_k\right)}{\sum_{v_p \in e_j} \exp\left(\mathbf{w}_{ah}^{\mathrm{T}} \mathbf{u}_p\right)},$$

$$\mathbf{u}_k = \mathrm{LeakyReLU}\left(\mathbf{W}_h \mathbf{h}_k^{l-1}\right),$$

2. Edge-Aware Node Update

$$\mathbf{h}_i^l = \mathrm{LeakyReLU}\left(\sum_{v_i \in e_k} \beta_{ij} \mathbf{W}_e \mathbf{g}_k^l\right),$$

$$\beta_{ki} = \frac{\exp\left(\mathbf{w}_{ae}^{\mathrm{T}} \mathbf{z}_k\right)}{\sum_{v_i \in e_q} \exp\left(\mathbf{w}_{ae}^{\mathrm{T}} \mathbf{z}_i\right)},$$

$$\mathbf{z}_k = \mathrm{LeakyReLU}\left(\left[\mathbf{W}_e \mathbf{g}_k^l \| \mathbf{W}_h \mathbf{h}_i^{l-1}\right]\right),$$

Experiments

Evaluate metric: F1-Macro on datasets PE and CDCP.

Model	ACTC	ARI	ARTC
Deep-Res-LG	65.3	-	15.0
St-SVM-strict	73.2	-	-
TSP-PLBA	78.9	-	18.7
BERT-Trans	82.5	**67.8**	**-**
EEGCN	**84.1**	60.3	***67.5**

CDCP

Model	ACTC	ARI	ARTC
Joint-ILP	82.6	75.1	68.1
Joint-PN	84.9	76.7	-
LSTM+dist	87.3	80.7	79.0
BERT-Trans	88.4	82.5	**81.0**
HGTrans	**89.5**	**83.1**	77.6

PE

Discussion

We address the Argumentation Mining task through a systematic approach that involves the construction of a meticulously designed graph, followed by the utilization of Graph Neural Networks (GNN) to extract meaningful representations from the graph. This transformation effectively converts the original task into node classification and link prediction tasks.

However, this approach may be perceived as redundant and less sophisticated, diminishing its appeal. Critics may argue, 'Why bother constructing a graph when you can simply input the text into ChatGPT, which possesses inherent knowledge of textual relations?'

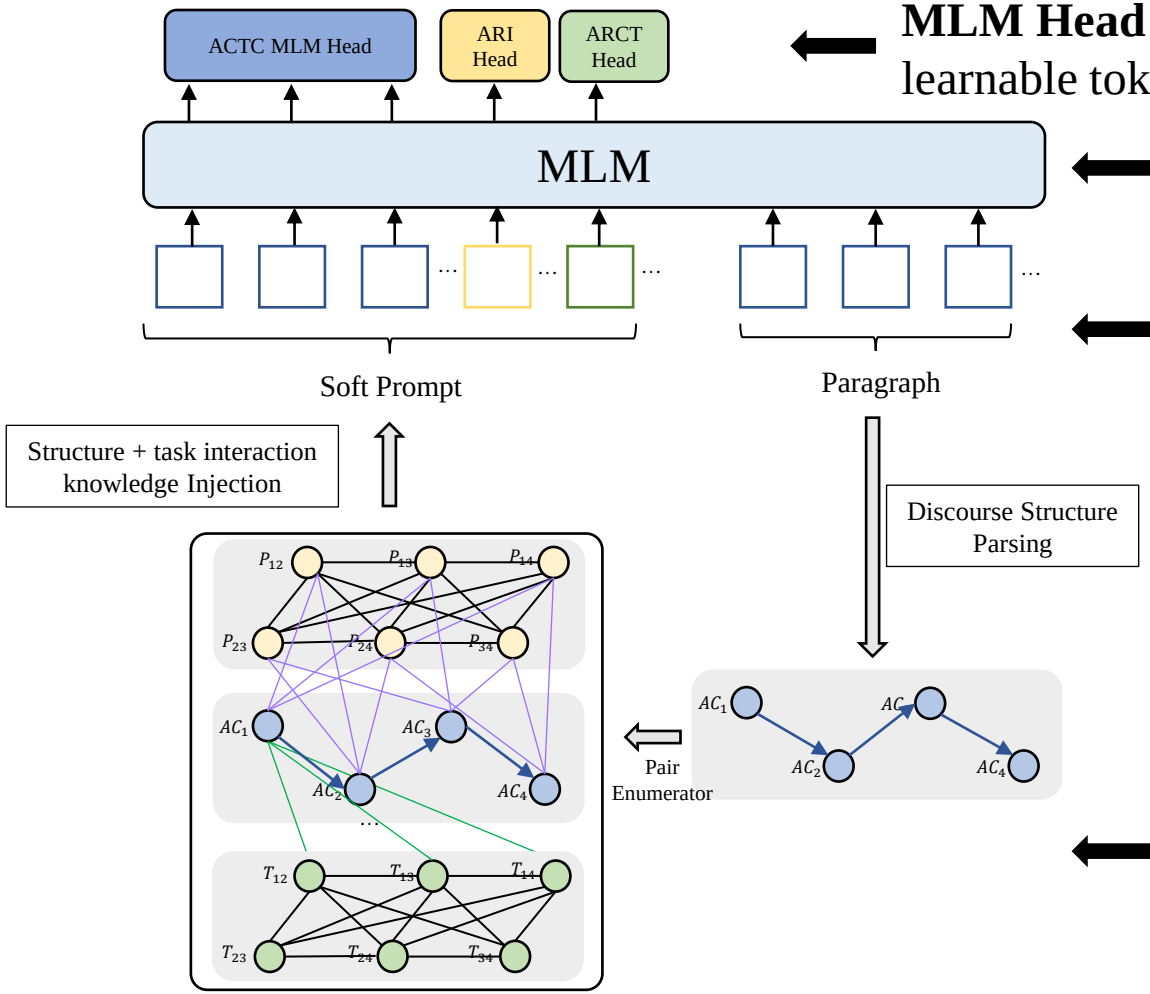
To address these concerns, we propose an alternative methodology leveraging a soft prompt scheme to leverage the capabilities of the language model. Empirical evaluations consistently demonstrate superior performance compared to conventional methods.

Motivation

1. Bridging the gap between pre-training tasks and argumentation analysis tasks.
2. Previous methods decode tasks in a specific order, first classifying components and then identifying and classifying relationships between component pairs. This results in an imbalance in feature interactions between tasks, where subsequently extracted features lack direct connections to previous ones. Therefore, we believe it is necessary to establish interactions among multiple tasks, which is crucial for multitask argumentation. For example, a classified argumentative component pair can infer a high probability of the component pair belonging to argument and evidence, and there is likely a relationship between the classified argumentative component pairs.
3. Previous methods often utilize recurrent neural networks or pretrained language models to model the semantic representation of argumentative components for argument mining. However, they overlook the discourse structure relationships between argumentative components (or other relationships) that are closely tied to the argumentative structure within the argumentative text.

Methods

The prior label information denotes the proportion of different labels in train set multiplied by the label word representations.



MLM Head Knowledge Injection: labels from three tasks are learnable tokens and initialized by the prior label information.

Mask Strategy: [AC], [P] and [T] only focus on its corresponding AC segments and ignore other parts. [D] can see all AC segments.

Soft Prompt Format:

$$[D][AC_1][mask][AC_2] \dots [AC_n][mask][P_{1,2}][mask] \dots [P_{n,n-1}][mask][T_{1,2}][mask] \dots [T_{n,n-1}][mask]$$

Where [D],[AC], [P] and [T] denote the document, argument, argument pair relation, argument pair relation type, which are learnable tokens and initialized by the corresponding prior label information.

Structure + Task Interaction Knowledge Injection: we inject the discourse structural knowledge and prior task interaction information into tokens by updating the token representations with RGCN.